



COURSE: AEROSPACE MATERIALS I (251-15333)			
DEGREE: BACHELOR IN AEROSPACE ENGINEERING		YEAR: 2	TERM: 1st
CLASSES G45: Wed 15:30-17:00 h (7.1.J06) and Fri 15:30-17:00 h (7.1.J03) Lecturer: Dr. Sophia A. Tsipas G46: Wed 17:15-18:45 h (7.2.J02) and Fri 17:15-18:45 h (7.2.J02) Lecturer: Dr. Juan Alberto Meza			

WEEKLY PLANNING				
WEEK	DATE	SESSION	TOPIC	DESCRIPTION
1	9 Sept	1	PRESENTATION OF THE COURSE. TOPIC 1. BONDING IN SOLIDS	Introduction to the course. Course structure and Evaluation. Bonding in solids. Relationship between bonding, structure and properties of materials
	11 Sept	2	TOPIC 2. STRUCTURE OF MATERIALS I	Crystalline systems and some definitions. Important metallic structures. Atomic positions. Directions in the crystalline cells. Planes in the crystalline cells
2	16 Sept	3	TOPIC 2. STRUCTURE OF MATERIALS II	Calculation of the atomic density. Crystalline defects. Solid solutions. Polymorphism and Allotropy. Amorphous Materials. PROBLEMS
	18 Sept	4	TOPIC 2. STRUCTURE OF MATERIALS III	RESOLUTION OF PROBLEMS
3	23 Sept	5	TOPIC 3. DIFFUSION IN SOLIDS. MASS TRANSPORT	Definition of diffusion. Diffusion mechanisms. Fick's diffusion laws. Diffusivity. Industrial applications. Examples.
	25 Sept	6	TOPIC 4. PHASE DIAGRAMS I	Basic concepts: Component, phase, micro constituent. One component phase diagrams. Binary isomorphous phase diagrams. Gibbs Phase Rule. Tie line and lever rule. Non equilibrium solidification. Binary eutectic systems.
4	30 Sept	7	TEST 1 AND PROBLEMS ON TOPIC 4	TEST 1 (Topics 1, 2, and 3). Resolution of PROBLEMS
	2 Oct	8	TOPIC 4. PHASE DIAGRAMS II	Precipitation in solid state. Invariant reactions. Intermetallic components. Congruent and incongruent melting. PROBLEMS
5	7 Oct	9	TOPIC 4. PHASE DIAGRAMS III	Fe-C System. Equilibrium transformations in the metastable Fe-C system. Eutectoid steels. Pearlitic transformations. Hypoeutectoid steels. Hypereutectoid steels. Influence of alloy elements in metastable Fe-C. PROBLEMS
	9 Oct	10	Bank Holiday	Bank Holiday
6	14 Oct	11	No lecture	Lecture will be rescheduled
	16 Oct		TOPIC 4. PHASE DIAGRAMS IV	Ceramic phase diagrams. Ternary phase diagrams. Ternary phase diagrams in metallic alloys. Ternary phase diagrams in ceramics. Resolution of PROBLEMS.
7	21 Oct	12	TOPIC 5. MECHANICAL PROPERTIES	Basic Concepts. Uniaxial Tension Test. Stress-strain Curve: Elastic Zone. Stress-Strain Curve: Plastic Zone. Slip Systems Stress-strain Curve: Calculations. Ductility concepts. True Stress-strain Curves.
	23 Oct	13	TOPIC 5. MECHANICAL PROPERTIES	Strengthening Mechanisms. Hardness. PROBLEMS.
8	28 Oct	14	TOPIC 6. ELECTRICAL PROPERTIES	Classic Model. Classification. Diagrams of Energy-Bands. Conducting materials. Metallic conductors. Ceramic conductors. Semiconducting materials. Intrinsic. Extrinsic. Insulating materials.

	30 Oct	15	TOPIC 7. MAGNETIC AND THERMAL PROPERTIES	MAGNETIC PROPERTIES. General concepts. Origin of the magnetic behaviour of materials. Types of magnetism. ferromagnetic, ferrimagnetic; paramagnetic; diamagnetic; anti-ferromagnetic. Magnetic domains. Hysteresis cycles. Types of materials according to their hysteresis cycles. Factors affecting the hysteresis cycle. Applications. THERMAL PROPERTIES.
9	4 Nov	16	TOPIC 8 CERAMIC MATERIALS I	Introduction. Structure of Ceramic Materials. Glasses. Mechanical Properties of Ceramic Materials.
	6 Nov		TOPIC 8 CERAMIC MATERIALS II	Processing of ceramic materials. Applications. PROBLEMS.
10	11 Nov	17	TEST 2 AND PROBLEMS	TEST 2 (Topics 4, 5, 6 and 7) Resolution of PROBLEMS.
	13 Nov	18	TOPIC 8 CERAMIC MATERIALS I	PROBLEMS ON TOPICS 5,6,7,8
11	18 Nov	19	TOPIC 9 POLYMERS I	Introduction. General overview. Historical development. Basics. Chemical structure of polymers. Nomenclature. Size and shape of polymers. Molecular weight and its distribution. Solid state of polymers. Crystalline polymers. Thermal transitions. Mechanical properties of polymers Viscoelasticity
	20 Nov	20	TOPIC 9 POLYMERS II	Processing and practical applications of polymeric materials PROBLEMS.
12	25 Nov	21	TOPIC 10 COMPOSITES I	Concept of composite materials. Constituents of composite materials. Classification. Why polymers are used in Composite Materials? Reinforcements. Particles. Composite materials reinforced with large particles. Fibres and prepregs. Mechanical properties of different reinforcement materials
	27 Nov	22	TOPIC 10 COMPOSITES II	Polymer matrix in Composite Materials. Thermoset and thermoplastic matrices. Epoxy matrix. Epoxy/amine network formation. Additives used to modify resin properties. Thermoplastic matrices. The interface region. Mean elastic properties of composites. Direction relative to a composite with and uniformly dispersed aggregate
13	2 Dec	23	TOPIC 10 COMPOSITES III	Fabrication Processes. Composite materials and Aerospace Construction Certification requirements for composites structures.
	4 Dec	24	TOPIC 11. ADHESIVES	Mechanisms of Adhesion. Modes of failure. Types of adhesives. Processing and design considerations. Testing methods and degradation mechanisms of adhesives. PROBLEMS
14	9 Dec	25	PROBLEMS ON TOPICS 8, 9 AND 10	Resolution of PROBLEMS
	11 Dic	26	TEST 3	TEST 3 (Topics 8, 9, 10 and 11)
LABORATORY SESSIONS: 1.0 A12				
	G46: 28/29 Oct	27	LABORATORY SESSION I	CRYSTALLINE STRUCTURES
	G46: 28/29 Oct	28	LABORATORY SESSION II	TENSILE TEST
	G46: 25/26 Nov	29	LABORATORY SESSION III	COMPOSITE MATERIALS
	G46: 25/26 Nov	30	LABORATORY SESSION IV	IDENTIFICATION OF POLYMERS